

CONCEPTUAL ROVER DESIGN FOR FUTURE TURKISH LUNAR MISSION

Beste Boybaşı* B. Teoman Kösoğlu†, Nurdeniz Altuntaş‡, Öznur Köşker§ Burak Yağlıoğlu¶
Koç University Istanbul Technical University TUBITAK UZAY
Istanbul, Türkiye Istanbul, Türkiye Ankara, Türkiye

ABSTRACT

Moon missions are growing in popularity, with numerous nations such as Russia, China, and the USA, and commercial enterprises aiming to explore the lunar surface. Türkiye is also taking a step to become one of the pioneers in the space race, with the "National Space Program", introduced in 2020 by the Turkish Space Agency (TUA). One of the main pillars of the National Space Program is the Moon Research Program where two missions are planned with targets of reaching the Lunar orbit and then the surface through hard landing within a few years (AYAP-1) and delivering a rover to the surface through soft landing (AYAP-2). In this manner, a group of AYAP-1 scholarship students of TUBITAK UZAY - Space Technologies Research Institute and Space Exploration Society (UKET) members has performed a conceptual design for a small Lunar rover considering AYAP-2 (soft landing mission). UKET is the main organizer of the international Anatolian Rover Challenge, which is the first and only international rover challenge in Türkiye and it aims to cultivate competent engineers for the space industry for TUA and TUBITAK UZAY. In this paper, the design factors of the rover are discussed, including the choice of the instruments, the rover's size and mobility, payload capacity, subsystems, the technical budgets needed for development and operations, and the conceptual design of the rover itself. The main goal in designing the rover is to carry out scientific research focused on the physical properties of the landing site, detect and analyze the concentrations of different minerals and elements, measure the growth of organisms under the conditions of the Moon, and investigate how the mining capacity of microorganisms is effected from the lunar conditions.

INTRODUCTION

The National Space Program of Turkey prepared by the Turkish Space Agency (TUA) was announced in February 2020. One of the main pillars of the National Space Program is the Moon Research Program, which will be conducted in two phases. The first phase (AYAP-1) features lunar orbital operations, hybrid propulsion system utilization, and hard landing on the surface [Baygeldi et al., 2023; Yağlıoğlu, et al., 2023]. The second phase (AYAP-2), is aimed to deliver a rover to the lunar surface through a soft landing [Baygeldi et al., 2023; Yağlıoğlu, et al., 2023]. Türkiye's

*BE Student in Mechanical Engineering, Email: beste.boybasi@tubitak.gov.tr

†BE Student in Electrical Engineering, Email: barbaros.kosoglu@tubitak.gov.tr

‡ME Student in Geophysical Engineering, Email: nurdeniz.altuntas@tubitak.gov.tr

§BS Student in Molecular Biology and Genetics, Email: oznur.kosker@tubitak.gov.tr

¶Chief Researcher at TUBITAK UZAY, Email: burak.yaglioglu@tubitak.gov.tr

motivation is to improve its space technology capabilities, gather deep space heritage for its domestic systems, foster international collaborations in terms of space science and technology, and develop an awareness of space technologies inside the society. TUBITAK UZAY is responsible for the design, development, and realization of the first Turkish Lunar mission.

The utilization of lunar rovers started in 1970 with the Lunokhod-1 rover, which is the first mobile research robot on the lunar surface. The main aim of the first rover mission was to analyze the lunar regolith and topography. It transferred more than 20000 images and conducted 25 soil analyses [Siddiqi, 2018]. During the Apollo missions in the early 70s, the lunar rovers were developed for the mobility of astronauts as well as for transporting equipment and lunar samples. In later years, the focus of lunar missions shifted to Mars, however, lunar rovers regained popularity with the Chinese Lunar Exploration Program. The Chinese have so far carried out 2 lander-rover missions called Chang'e 3 and Chang'e 4 and rovers were called Yutu-1 and Yutu-2. Yutu-1 is a 6-wheel autonomous rover with around 140 kg, served almost 3 years performing mineral and topographic analysis, astronomical observations, and image transfer, starting in 2013 [Sun et al., 2013]. Yutu-2 which landed on the far side of the moon in 2019 still continues its operations by analyzing topography, mineral composition, and solar winds [Qiong et al., 2016]. In addition to government missions, private companies also joined the space race, advancing commercial space applications. This situation allowed cost-effective and smaller-size rover missions with only 30 kg mass and sometimes also utilizing new technologies such as 3D printing technology [Shahrubudin et al., 2019]. One of the examples of private companies is Japan-based iSpace's Sorato rover. It originated from Google's Lunar XPRIZE mission "Team Hakuto". Sorato is a 4-wheeled, 3.8 kg rover to capture 3D images and analyzes obstacles [Walker, J., 2018]. Considering the mission plans as well as size, and mass constraints, the best reference technology for the design case is determined as NASA's Sojourner Rover. Sojourner weighs only 10 kg and was developed in 3 years [Bogue, 2012]. It is covered with solar panels, that are measured at 68x48x28 cm when fully deployed, and has a rocker-bogie suspension with 6 wheels. During its 83-day mission, Sojourner took pictures of Mars and made chemical, atmospheric, and topographic analyses [Siddiqi, 2018].

Along with the exploration of space, utilizing space resources will be a new achievement for humanity. The increasing need for metals used in technology brings the requirement of space mining. Among the mining technologies, the use of microorganisms for in-situ resource utilization (ISRU) is called biomineralization which has the potential for acquiring oxygen, water, and essential elements for self-sustaining human foundations. On the Moon, recovering and obtaining precious metals from the mineral-bearing regolith or rock can be performed [Gumulya et al., 2022; Santomartino et al., 2023].

In terms of the scientific payload of the Turkish Lunar Rover, this paper proposes a biomineralization strategy for metal utilization that is available on the Moon's surface. Considering the biology and geology experiments performed on previous rovers, space biomineralization experiments are not performed previously onboard the rovers. Our experimental design is created by taking the European Space Agency BioRock experiment as a reference, which is performed at International Space Station. In BioRock, the bioleaching of rare earth elements from Moon and Mars analog basalt is explored by three bacterial species [Cockell et al., 2020]. Similarly, two species are chosen in our design for investigating their mining potential of iron oxide, thorium, and uranium which could be potential targets for their economic value. In order to extend the research on space biomineralization, our science experimental design will focus on biomineralization experiments.

In this paper, a conceptual rover design is proposed for the second phase of the Turkish Lunar Mission that is planned within a decade. The sub-systems of the designed rover include navigation and mobility, data handling and communication, power, thermal, structure and mechanisms, and

science module. This paper will explain the general features of the conceptual rover mission and rover design as well as the science objectives, payloads, and subsystems.

LANDING SITE AND MISSION DESCRIPTION

The Copernicus Crater

The Copernicus Crater, one of the significant Copernican impact craters that result from a large meteor impact, is located at the near-side of the Moon (9.6°N, 340°E). The Copernicus Crater is very important for the understanding of the dynamics, and geological timescale of the Moon. Because thanks to large impact craters, scientists can define the lunar time and stratigraphic units with events and their deposits [Oberbeck, 1975]. The Copernicus Crater is one of them, as it results from a large meteor impact and its diameter is approximately 93 km. Its well-preserved features, mineral and element compositions, ray system, deposits, and central peaks are very important for scientific research. Also, for Turkish Lunar Rover mission, it has an important role due to its mineral and element composition since biomining is the primary objective of this mission.

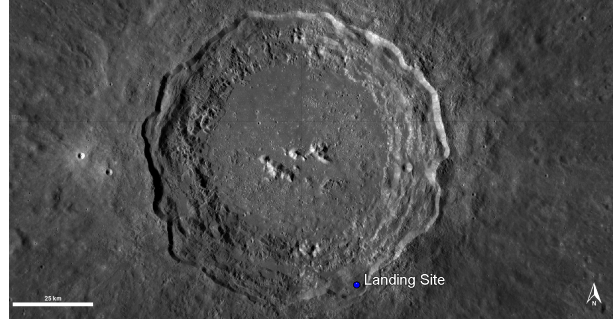


Figure 1: Copernicus Crater and Landing Site

Lunar crust is dominated by three major geological terranes which are the Procellarum KREEP Terrane (PKT), the Feldspathic Highlands Terrane (FHT), and the South Pole Aitken Terrane (SPAT) [Jolliff et al., 2000]. PKT's KREEP represents (K) Potassium, (REE) Rare Earth Elements and (P) Phosphorus. Furthermore, the PKT region (mare and non-mare) is rich in terms of FeO (iron(II)oxide) (%17.3) (Figure 3a), Th (thorium) (5 ppm) (Figure 3b), olivine (mostly magnesian olivine) (Figure 2), plagioclase (Figure 2), pyroxene (Figure 2), U (uranium) (2.1 ppm) (Figure 3c) [Arai et al., 2011; Jolliff et al., 2000; Lucey et al., 1991; Yamashita et al., 2010]. PKT terrane includes Oceanus Procellarum, Mare Imbrium, and the Copernicus Crater, which is selected as a soft landing site for the Turkish Lunar Mission (Figure 1).

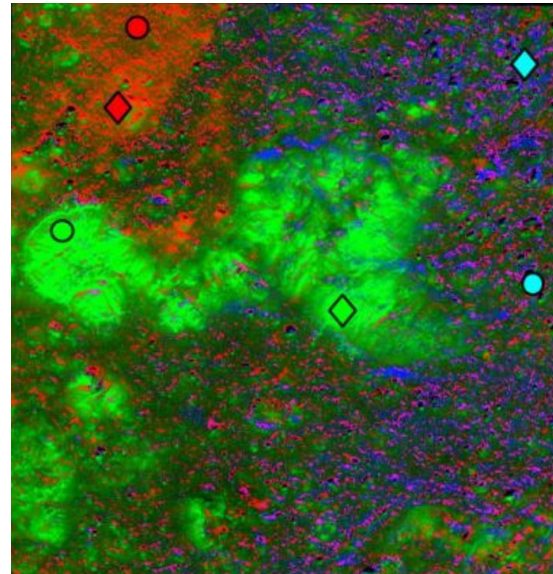


Figure 2: Olivine, pyroxene, and plagioclase-rich areas in the central peak of the Copernicus Crater. Green represent olivine-rich areas, red represents pyroxene-rich areas, and blue represents plagioclase-rich areas. [Arai et al., 2011]

Figure 3a and Figure 3b show the concentrations of FeO, Th in the Copernicus Crater, and Figure 3c shows U in the Moon [Yamashita et al., 2010]. Inside the Copernicus Crater, these elements' concentration is weaker than the outside of the crater. Erosion after the impact, mixing surface and deep-layer materials, and scattering all the materials after the impact could be reasons for this difference. Based on the different densities of elements inside and outside the crater, the most

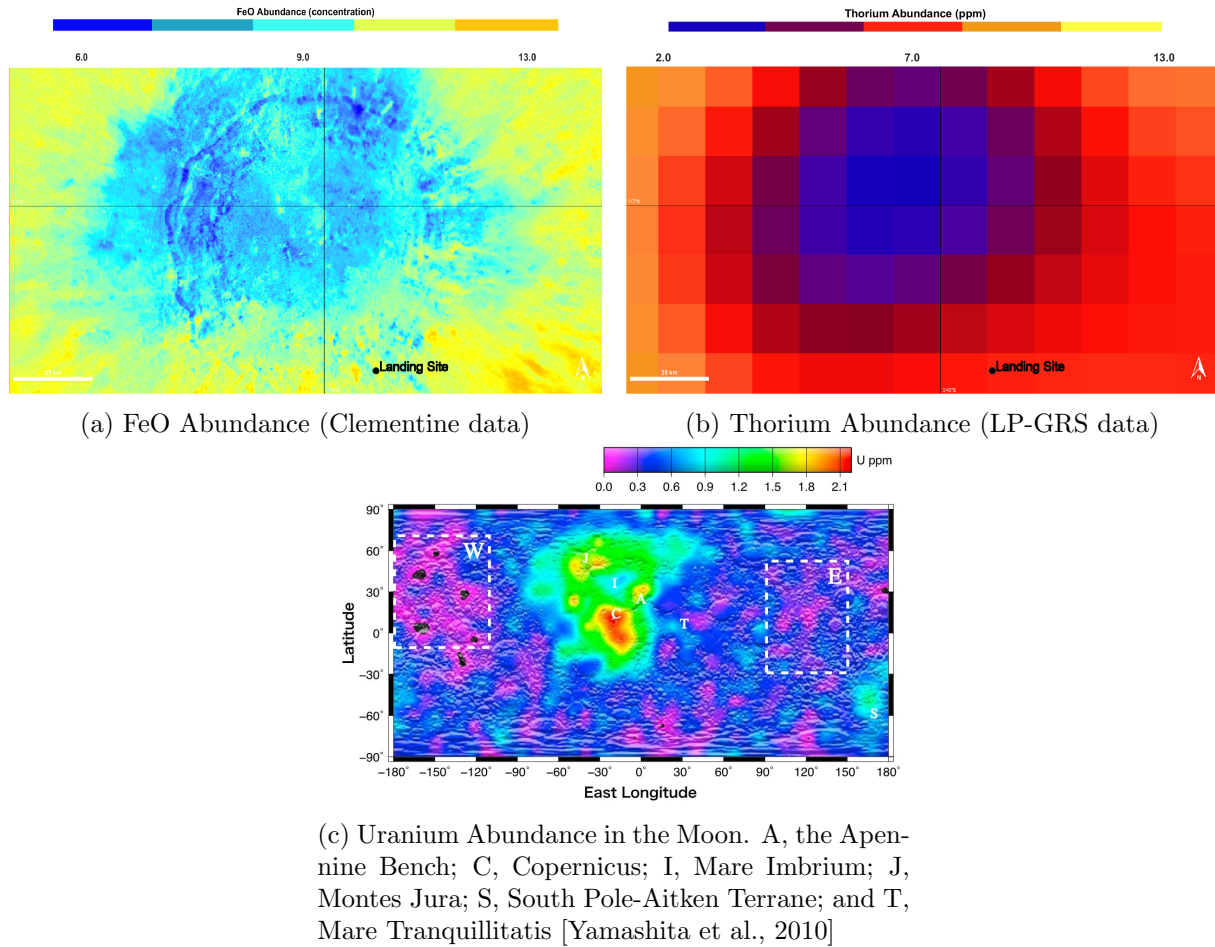


Figure 3: FeO, Th and U abundance maps in the Moon surface

suitable area is selected for the landing site. It is decided that the flat area located on the southeast wall of the Copernicus Crater is the most appropriate site for landing, both for a soft landing and due to the presence of specific elements like FeO, Thorium, and Uranium, in that area (Figure 3). These specific elements are important for scientific experiments.

Description of the Mission

The primary objective of this mission is to biomine the FeO, Th, and U elements on the Lunar surface. The secondary objective of this mission is to detect FeO, Th, U, olivine, pyroxene, plagioclase, etc., and analyze their concentration in the southeast wall of the Copernicus Crater. These elements are essential since they have great potential for generating self-sustaining extraterrestrial colonies. To achieve these objectives, the rover will be equipped with a sample collecting mechanism, science module and biosensor device, cameras, sensors, and instruments such as X-ray Fluorescence (XRF) spectrometer, ultraviolet-visible (UV-VIS) spectrophotometer, and dosimeter. The collected data from these instruments will be utilized to develop graphs and maps of the concentration of the elements and minerals, obtain the biomining experiment results, and to create a map and model of the radiation environment in the wall of the crater. This information can be used to better understand the potential opportunities, risks, and challenges for future lunar exploration, utilization and habitation, as well as to develop mitigation strategies to protect astronauts and infrastructure.

Trajectory for Lunar Rover

The mission will last for half a Lunar day (10-12 Earth days) in the daytime. The southeast wall of the Copernicus Crater is considered a landing site. Figure 4 shows the planned mission profile, landing site, and path of the rover on the southeast wall. The lunar rover is expected to travel within

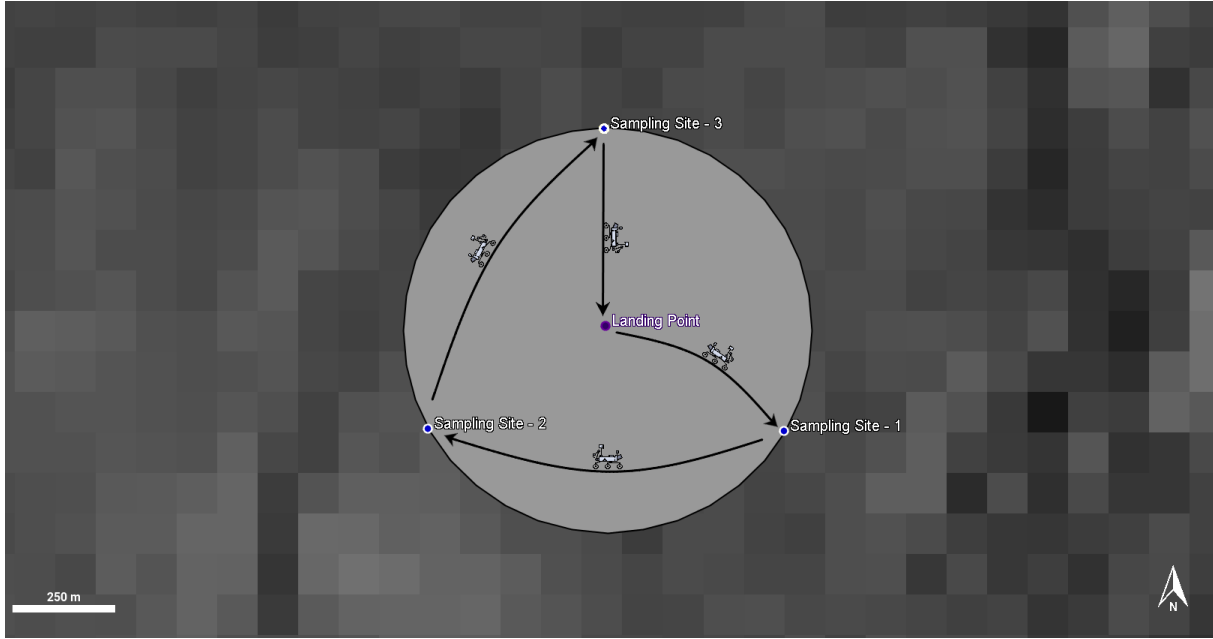


Figure 4: Turkish Lunar Rover's trajectory in the Copernicus Crater's southeast wall. The purple dot represents the landing point, arrows represent the path of the rover. Sampling sites are also represented as blue dots.

a circle with a radius of 0.5 km and collect samples inside this circle. The collection of samples will be completed within two days when the speed limitation of the rover is considered. After landing at the center point (8.314°N, 340.566°E), the rover has to go to sampling sites by order: Sampling Site-1 (8.278°N, 340.54°E), Sampling Site-2 (8.257°N, 340.513°E), Sampling Site-3 (8.271°N, 340.525°E). Between these missions, the rover is expected to stop and take panoramic photographs of the site. After collecting samples from these three sites, the rover's final destination is to return back to the Landing Point (8.314°N, 340.566°E). Returning back to the landing point may be necessary for lander resource utilization in terms of complex sample processing, extra power, or large-volume data transmission to the Earth through the lander. An additional reason for returning back to the lander is utilizing its resources in terms of any occurrence of a problem in the rover, for instance, the power that is necessary for the scientific experiments can be met by the lander if the rover encounters a power issue.

PAYLOADS AND INSTRUMENTS

The current payload in the designed Turkish Lunar Rover includes X-ray fluorescence (XRF) spectrometer, ultraviolet-visible (UV-VIS) spectrophotometer, dosimeter, bioreactors, and cameras.

Dosimeter

Radiation is one of the main considerations in space flights due to its high damage to living cells and organisms [Reisz et al., 2014]. Therefore, measuring how much radiation the cells are exposed to is important in order to measure how their biomining capability gets affected and the vulnerability of cells to the radiation dose that they get during mission time. In general, the characterization of the radiation on the landing site will be helpful for future human flights. Therefore, the need for quantifying and mitigating exposure to damaging radiation is significant. A dosimeter will be used to characterize the ionizing radiation on the Copernicus Crater.

Bioreactor

For the investigation of the effects of Moon radiation and low gravity on the microorganisms that are able to perform biomining, miniature bioreactor devices will be utilized. Bioreactors will allow for

generating a proper setting for selected organisms to grow. They are generally used in biotechnology for the bulk production of industrially important microorganisms. They are adapted for space biomining in order to control and monitor cell cultures [Walther, 2002].

X-Ray Fluorescence Spectrometer (XRF)

X-ray fluorescence spectrometer (XRF) has been used for the determination of rare earth elements (REE) since the 1970s'. In this mission, energy-dispersive X-ray fluorescence (EDXRF) analysis is selected due to its size and simplicity rather than wavelength-dispersive X-ray analysis. In EDXRF, XRF radiation is detected by a semiconductor detector which could be a pin diode, silicon drift chamber, or silicon lithium semiconductor [Schramm, 2016]. XRF's basic methodology includes the following steps: An atom is excited with X-ray and then the atom transitions from the lower energy shell to the higher energy shell. When the atom turns back to its lower energy shell, it makes radiation called fluorescence. Each mineral and element has its fluorescence pattern. So with this methodology, minerals and elements can be recognized according to their patterns with XRF. Basically, XRF consists of X-ray tube, filter, detector, and sample (Figure 5). Thanks to the XRF, FeO, Th, U, olivine, pyroxene, and plagioclase can be detected in the mission. The elemental composition of the rock samples collected will be determined before the experiment starts, and the released elements by the leaching activity of biomining bacteria will be detected to measure the success of biomining.

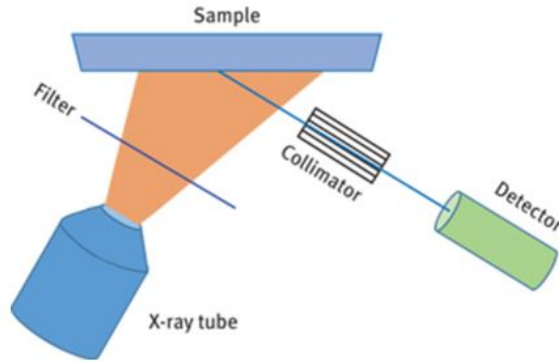


Figure 5: Schematic setup of an EDXRF [Schramm, 2016]

Ultraviolet-Visible Spectrophotometer (UV-VIS)

The UV-VIS spectrophotometer has been chosen for this mission due to its compact and simple design, as well as its suitability for measuring and tracking the growth of microorganisms from the optical density of microbial cultures. OD600 measurement will be performed to track the turbidity of solutions in bioreactors. UV-VIS's basic working method follows these steps: Firstly, both Tungsten and D2 lamps emit light. Tungsten emits visible light and D2 emits ultraviolet light. These lights reflected from the mirror as one, and come to the monochromator. A monochromator splits this light in a chosen wavelength. After that, the light with the chosen wavelength goes through both the reference and the sample. Finally, the intensity of the light that is absorbed in different wavelengths is measured with the detector (Figure 6) [Rocha et al., 2018]. By analyzing the differences in absorbance measured by the detector, the concentration of bacterial culture will be measured. When the light passes through the culture is lower, it means that the cell density is higher [Domańska et al., 2019].

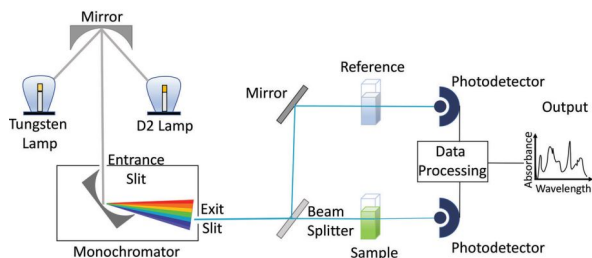


Figure 6: Schematic setup of a UV-VIS spectrophotometer. The deuterium lamp generates ultraviolet light (UV region) and the tungsten lamp emits visible light (VIS region) [Rocha et al., 2018].

IMAGERY

The requirements of the imaging for the rover mission are summarized as follows:

- **Dust and Radiation Protection:** Protective covers to shield cameras from abrasive lunar dust and radiation effects.
- **Energy Efficiency:** Energy-efficient cameras to maximize the rover's operational time.
- **Data Compression and Transmission:** Efficient data compression and reliable transmission protocols for limited data storage and bandwidth.
- **Redundancy:** Including redundant cameras to ensure uninterrupted functionality and data collection.

Camera Type	Purpose and Function
Panoramic Camera	High-resolution, wide-angle imaging of the lunar surface for comprehensive coverage
Microscopic Imager	Detailed examination of lunar soil, rocks, and potential biological traces
Stereo Navigation Cameras	Generation of 3D maps and aiding in rover navigation
Scientific Cameras	Aiding science experiment in rover

Table 1: Camera Types and Their Functions

When designing the camera section of the Turkish Lunar Rover, it is crucial to take into account the mission's scientific objectives and the unique challenges of the lunar environment. For comprehensive coverage of the lunar surface and to fulfill the diverse scientific objectives, the cameras on the lunar rover are strategically positioned at various locations. Panoramic cameras are situated at the rover's front to provide a wide-angle view, capturing high-resolution images of the terrain ahead. To facilitate a detailed examination of lunar soil and rocks, the microscopic imager is typically mounted at a lower height near the rover's front or side. Finally, the stereo navigation cameras are placed at the rover's front and rear, aiding in precise movement and generating 3D maps to assist in navigation through challenging lunar terrain. These well-placed cameras collectively enable the rover to carry out its scientific mission with efficiency and accuracy. To withstand the harsh lunar conditions, the cameras must have dust and radiation protection. Energy efficiency and data compression are essential to manage power and data storage limitations. By considering these aspects, the camera section of the lunar rover can play a pivotal role in achieving scientific goals and advancing our understanding of the Moon's geology and environment.

SCIENCE MODULE

The Moon's surface is a valuable scientific research field since it has low gravity, and it is a naturally radiation-exposed celestial body due to the absence of a magnetic field and atmosphere [de Vera et al., 2012; Horneck, 2011]. For scientists aiming to test the effects of low gravity and high radiation on organisms, the Moon's surface is a good candidate. Together with the requirement to learn the effects of space on biomineralization, several missions are performed. Up to now, Foton M4 - SCK CEN, ESA - BioRock, and BioAsteroid are the main space biomineralization experiments performed. Biomineralization is leaching and extracting valuable elements from rocks or regolith with several methods that mining bacteria use, such as weathering the rocks and solubilizing cations by using organic acids [Averesch et al., 2023; Biorock, 2023] Figure 7 shows several interactions of bacteria with metals, including bioleaching.

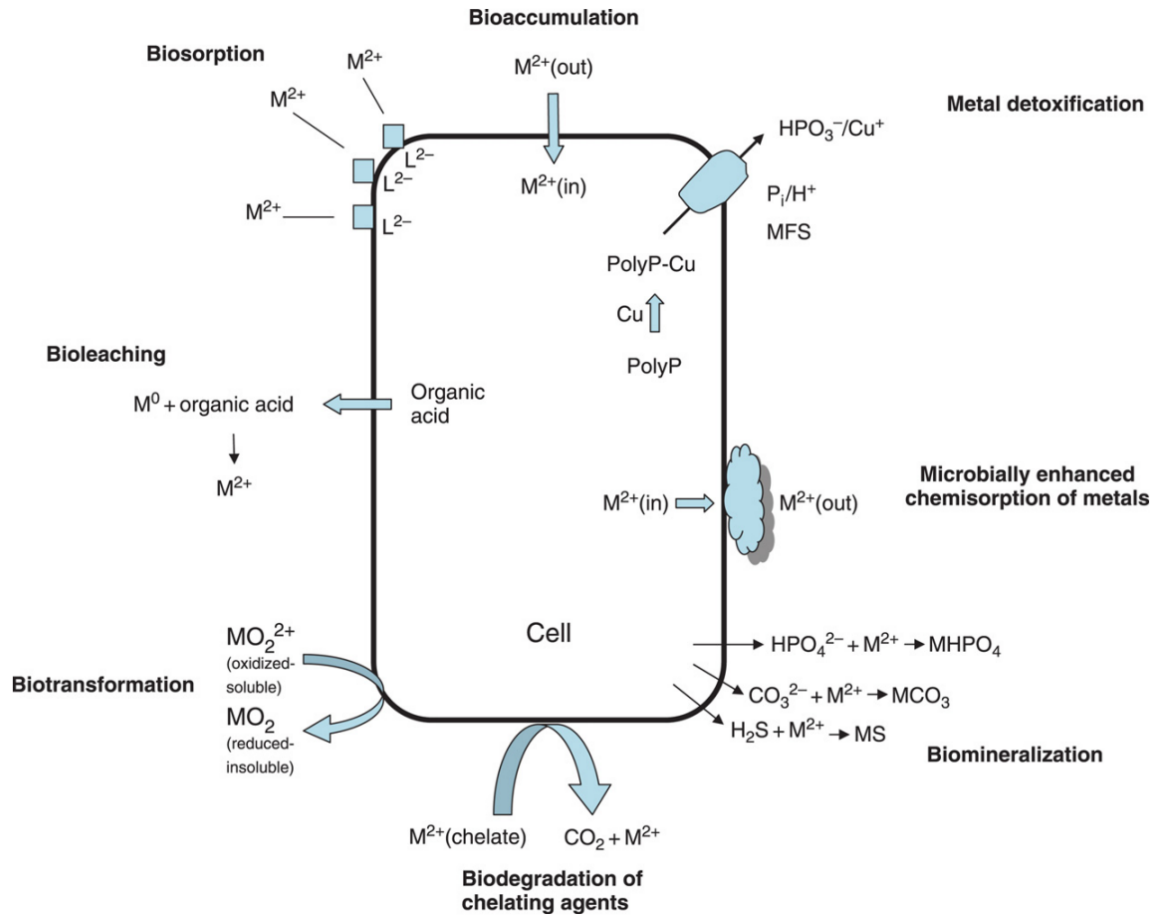


Figure 7: Bacteria-metal interactions. Bioleaching is described as the conversion of insoluble metals to soluble metals by microbial processes [Jerez, 2017].

In the BioRock experiment, the aim was to examine bacterial biofilm growth on rock and microbe-mineral interactions at the conditions of ISS. They utilized a heap-leaching biomining reactor that uses biofilm on rocks in order to extract metals [Loudon et al., 2017]. The reason for using biofilms for biomining is that they can accelerate the leaching of metals [Gumulya et al., 2022] and the fact that most leaching bacteria can grow by attaching to the surface of a solid substrate [Jerez, 2017]. *Sphingomonas desiccabilis*, *Bacillus subtilis* NCBIB 3610, and *Cupriavidus metallidurans* CH34 were the bacteria species studied since they are known to interact with rocks and minerals and can survive desiccation. Bioreactors were located inside a centrifuge to grow microorganisms in microgravity and simulated Martian gravity. At the end of the experiment, the biomining reactors are brought back to the Earth from ISS, and the elemental release rates, meaning the amount of leached metals, are investigated by ICP-MS/OES spectroscopy technique. [Loudon et al., 2017]. Briefly, the results of the experiment showed that in comparison to non-biological controls, *S. desiccabilis* had higher leached metal amounts of rare earth elements. In comparison to sterile controls, *S. desiccabilis* and *B. subtilis* increased vanadium leaching at three gravity settings [Gumulya et al., 2022].

The designed science mission for our conceptual rover mainly follows the principles of BioRock design. In our experiment that aims for extraterrestrial biomining, a miniature bioreactor that is similar to the bioreactor used in the BioRock experiment will be adjusted. The design and principle of the BioRock bioreactor are given in Figure 8. Our concept aims to design a rover that can reach the proper sampling destinations, carry the rock samples to the onboard biological experiment payload, and transmit the data gathered by cameras and spectrometers throughout the experiments. The effect of ionizing radiation and Moon gravity (0.17g) on the biomining potential and the viability of cells will be measured onboard the rover [Gumulya et al., 2022].

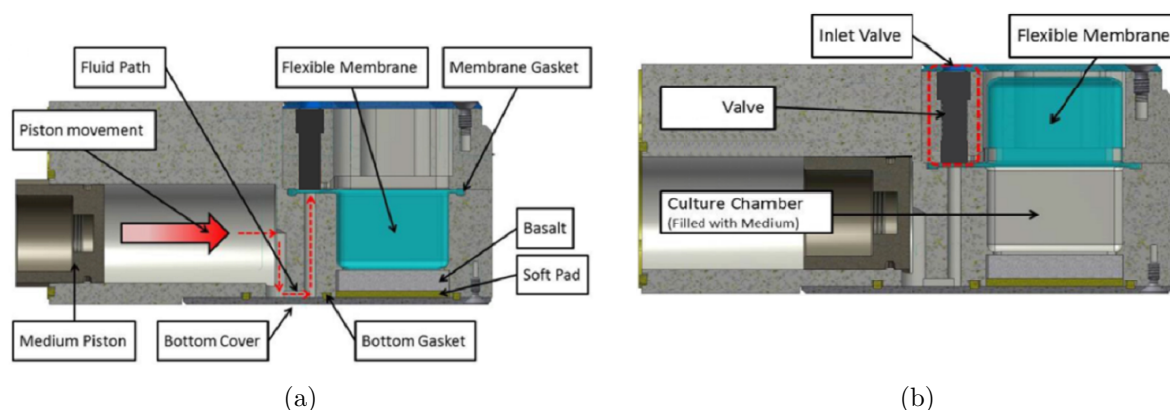


Figure 8: Schematic illustration of BioRock biomining reactor hardware. (a) Before the culture media is introduced. (b) After the culture media is introduced [Loudon et al., 2017].

Comparing the effects of the Moon's gravity and microgravity with Earth's gravity will be achieved by repeating the same experimental setup on Earth and ISS for all the bacteria species chosen since simulating microgravity on Earth with all aspects is not possible [Gumulya et al., 2022]. In that way, it will be possible to compare microbe-mineral interaction in three different gravity conditions. As a substrate, rock samples collected during the Apollo 12 mission will be utilized on Earth and ISS experiments since the landing site of Apollo 12 was close to the Copernicus crater and collected samples are from the rays of Copernicus crater [Korotev et al., 2011; Young, 2017]. Triple replicates will be performed for each gravity condition; Earth, Moon, and ISS microgravity by utilizing three miniature biomining reactors as replicates for each organism. Non-biological controls are utilized to control whether the dissolution of metals from rocks is performed by the bacteria or another abiotic factor. Figure 9 shows the replicate experiments for gravity conditions and microorganisms.

For the control of radiation, Earth and ISS experimental sets will be used. Measuring the effects of radiation on microorganisms on the Moon is important since most microbial experiments that take place in Low Earth Orbit are performed away from damaging radiation. Galactic cosmic rays and solar energetic particles are two radiation types that are harmful to microorganisms beyond the magnetosphere. Therefore, the Moon experiments that will be performed without the protection of the magnetosphere will measure the effect of deep space radiation [Reisz et al., 2014; Gumulya et al., 2022].

Our experiment will utilize two species of biomining bacteria for iron, thorium and uranium mining. *Acidithiobacillus ferrooxidans* is an acidophilic and chemolithotrophic biomining bacterium that can leach iron and uranium from ores. *A. ferrooxidans* is the model biomining bacteria, which is the most studied leaching bacteria species and is able to grow on biofilms as a monolayer [Jerez, 2017]. It makes a major contribution to the oxidation of Fe(II) to Fe(III) biologically in biogeochemical cycling. It has several iron uptake systems and can utilize iron as an electron source where iron is scarce [Lefimil et al., 2009; Quatrini et al., 2005]. Besides from its capability for the oxidation of ferrous iron to ferric iron, it can also leach uranium from ores. U^{4+} is converted to its soluble form U^{6+} by



Figure 9: Triplicate experiments to be performed for each gravity condition and each species.

using ferric iron (Fe^{3+}) as an electron acceptor [Pal et al., 2010; Abhilash et al., 2010].

The second biomining bacteria determined is *Pseudomonas aeruginosa*. *P. aeruginosa* has been shown to perform bioleaching from thorium-uranium concentrate and monazite, and it can leach these ores more effectively than abiotic leaching processes. In two studies examining the leached percentages of Thorium and Uranium, *P. aeruginosa* has been shown to leach them effectively. The percentage of Th-U concentrate and monazite leached by *P. aeruginosa* was observed as 42.1 and 52.6 respectively [Hassanien et al., 2013]. In another study examining the effects of bioproducts released from *P. aeruginosa* on the bioleaching of thorium (Th^{4+}) and uranium (UO_2^{2+}) from thorium-uranium concentrate, *P. aeruginosa* leached 68% of uranium and 65% of thorium. The siderophores it produces leached U and Th and create complexes with them. Siderophores are one of the secondary metabolites that have metal-chelating ability, and they can scavenge Fe^{3+} valuable metals in the extracellular environment. *P. aeruginosa* is one of the most common bacteria that can produce siderophores [Desouky et al., 2016]. These two species are convenient for bioleaching processes because of their resistance to metals and capability to perform metal absorption [Mathiyazhagan, 2011] that is why these two species are utilized in our experiment.

Two experimental setups are designed for *P. aeruginosa* and *A. ferrooxidans* inside different bioreactors. Sample treatment and analysis will be performed in situ by these miniature bioreactors and spectrometers on the rover. Investigation of the rock samples that will be collected within a circle with a radius of 500 m from the lander by drilling mechanism and Fe, Th, and U biomining potential of the selected microorganisms will be measured.

Bacteria cultures will be grown until they reach a certain optical density ($\text{OD}_{600}=1$) before their placement in the bioreactors. The principle of measuring optical density by UV-VIS spectrophotometer is measuring the light amount passing through the sample, the higher the bacteria concentration, the less light passes through [Sezonov et al., 2007]. After the measurement of optical density, bacteria species will be dehydrated and sent to the Moon in a dormant state. When a proper optical density is reached for all bacteria, which is the same for all to compare from the same starting point, how much they will be able to proliferate upon inoculation onto the Moon rock samples will be measured properly.

For the comparison of growth on the collected rock samples, the surface area of rock samples should be calculated, and relative growth should be measured. The surface area calculation of the rock samples to be delivered to the sample processing unit will be performed by using image processing algorithms. Two cameras that are located on top of the sample delivery platform will be used for generating 3D images to measure the surface area of the collected rock samples. Before the samples are delivered inside the bioreactor, their surface area will be determined with the image processing algorithm. Our drill mechanism will bring the rock samples to a platform supported by a piston. Surface area calculation will be performed, the rock sample will be delivered from the platform to the bioreactor with the piston mechanism, and then the sample chamber will be sealed for the experiment.

The bioreactors will contain three separate chambers. Two of them will include culturing media and dehydrated bacteria that are prepared and placed inside the related chambers of the bioreactor before launch. The third chamber is a transparent sample chamber where the rock sample will be delivered. The bacteria species will be active when the lid of the culture media and bacteria chambers are open and mixed together with the rock sample. The temperature inside the bioreactor is kept at a suitable temperature for each bacteria species. Each bioreactor will be adjusted to the proper environmental conditions in which the included bacteria species can live.

When the rock samples are delivered to the sample chamber, which is the transparent chamber, the chambers containing culture media and bacteria will be opened, and the bacteria will be rehydrated.

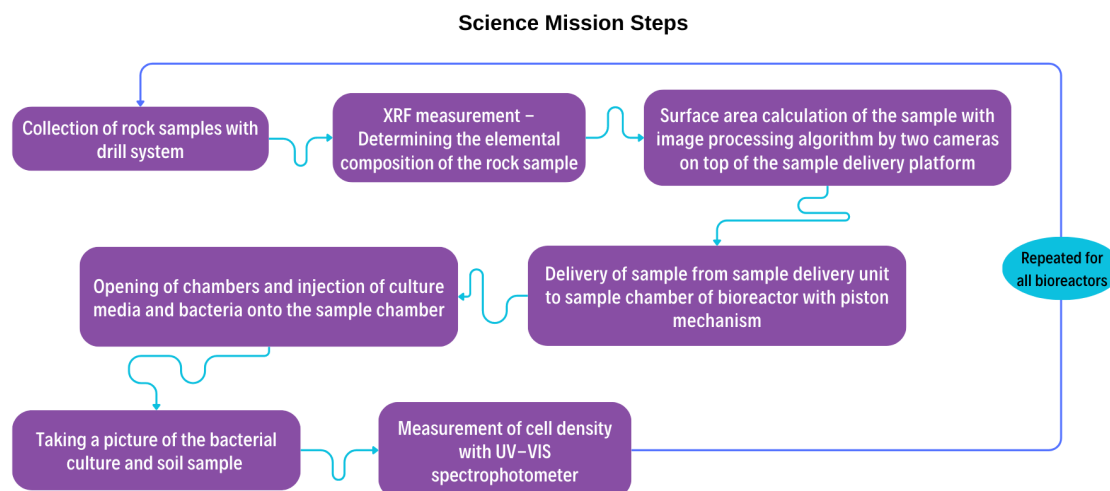


Figure 10: Triplicate experiments will be performed for each gravity condition and each species.

A camera will be placed inside the bioreactor on the opposite side of the sample unit where the rock sample will be located. A photograph of the sample rock will be taken before the start of the bio-mining experiment, during the period of bacterial biofilm growth, and at the end of the experiment after 8-10 Earth days. The first picture will be taken when bacteria, culture media, and sample are mixed as soon as the experiment starts in each bioreactor. The measurement of bacterial growth will be performed by UV-VIS spectrophotometer at OD(600) nm from the turbidity measurement [Li et al., 2023]. The turbidity measurement will be performed at certain intervals to track and compare bacterial growth. In order to validate whether selected bacteria species performed biomining, an X-ray fluorescence spectrometer (XRF) will be used to detect the elements and minerals given to the solution [Dhara & Misra, 2019]. A rotary platform system will be utilized in order to carry bioreactors to the sample delivery platform, to the UV-VIS spectrophotometer, and to the XRF when they are needed to be used. The steps of the science mission are explained in Figure 10 as a diagram.

Concisely, our science module will include two biomining bacteria that will be tested for their leaching ability of Fe, Th and U on the rock samples collected from the southeast wall of the Copernicus Crater. As one of the goals of space biology in resource utilization, harnessing minerals by using microorganisms onboard a rover will bring a new aspect to space biology. Our experimental design might potentially be valuable for moving space biomining one step further.

SAMPLE COLLECTING MECHANISM

The lunar surface is covered with a regolith layer, which means it is covered with unconsolidated and loose rock material. It contains small rock fragments, minerals, and elements and carries valuable information about the geological heritage of the Moon. To detect and collect the regolith samples for the biomining experiment, a sample-collecting mechanism is developed according to the regolith structure. Here, the first robotic drilling and coring device in the history of lunar exploration Apollo 11 and Luna 16 [Tang et al., 2016] were considered for reference. With similar methodology, a lunar drilling mechanism is considered to get the samples for the Turkish Lunar Rover.

Lunar drill mechanisms are more advantageous when compared to robotic arm graspers due to their placement and compact size. Since the drill mechanism is located under the rover in most cases, it is sheltered and therefore less exposed to the Sun's radiation. However, the heat generated during the drilling process requires thermal management for a successful mission. Also, drilling mechanisms require less degree of freedom in motion by enabling less complicated mechanisms especially compared to robotic arms.

In the conceptual design of the Turkish Lunar Rover, three different sample locations across the Copernicus Crater are determined to conduct three sets of experiments. In each location, the rover is designed for using the drill mechanism to collect regolith sample, which consists of mostly small rocks.

For the mechanism, the flexible tube coring principle is considered due to its applicability. The principle includes a flexible tube, core tube, drill pipe, and sealing device. Initially, the drill pipe houses both a flexible tube and a core tube. While rotating and moving downward, the flexible tube outside the core tube descends and curves inward to the bottom of the structure, encompassing the soil core until reaching the desired depth. At the end of the procedure, the soil core is present inside the core tube [Tang et. al., 2022]. Finally, the sealed tube is pulled out by the transferring mechanism. With this process, it is planned to drill to a depth of approximately 30 cm. The whole process can be visualized in Figure 11.

Besides the advantages mentioned above, the flexible tube coring principle also has various disadvantages. Firstly, using two nested tubes creates relative friction between their surface. This friction may result in a dragging force during the drilling. Moreover, the soil texture can affect the drilling quality significantly. In terms of coring, the volume of the sample soil can affect the drilling parameters and coring quality. To reduce the risk of the challenges, a percussive mechanism can be integrated to improve drilling efficiency, as utilized in the Luna 24 mission. With the percussive mechanism, the threshold for the penetrating force can be extended efficiently [Tang et al., 2016].

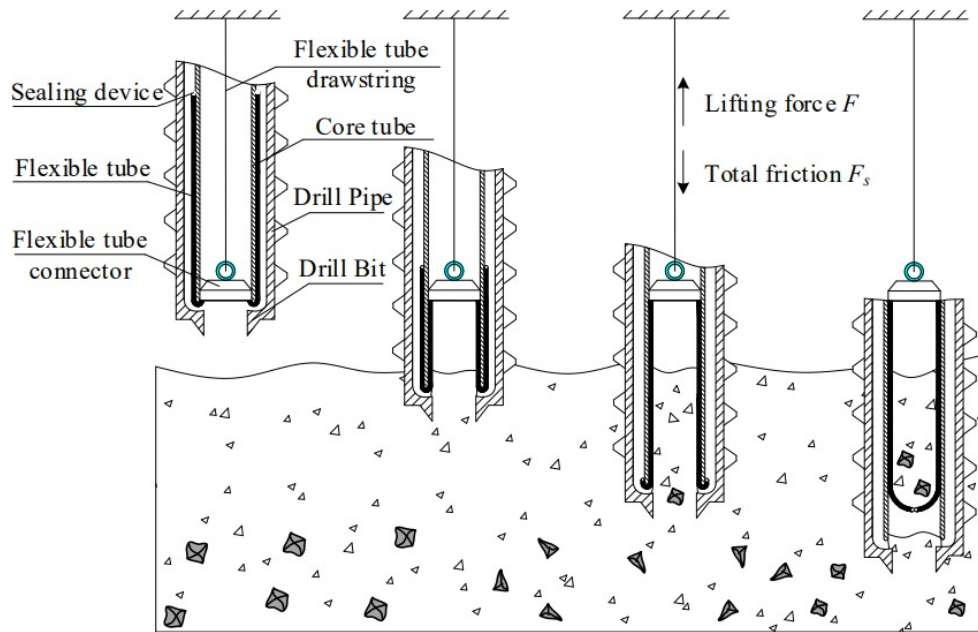


Figure 11: Flexible Tube Coring Principle [Tang et al., 2022]

Overall, to overcome all challenges, both the drilling and coring systems should be tested by building

a test stand. For better visualization, Table 2 shows the advantages and disadvantages of the flexible tube coring principle below.

Advantages	Disadvantages
Less degree of freedom	Dragging force between the nested tubes
Compact size compared to the graspers	Unpredictible soil texture

Table 2: Advantages and disadvantages of flexible tube coring principle

NAVIGATION AND MOBILITY SYSTEM

Navigation

The mission's navigation system for the lunar rover is a critical component that ensures accurate movement and positioning on the Moon's surface. This system may include a combination of advanced technologies such as a star tracker, inertial measurement unit (IMU), and terrain mapping cameras. An inertial measurement unit can detect the rover's acceleration and rotation, providing crucial data for precise navigation. A star tracker can be used to locate the rover by comparing the positions of stars in the sky with a known star map, enabling the rover to determine its exact position and orientation relative to the lunar surface. Additionally, terrain mapping cameras play a pivotal role in assisting the rover's navigation. These cameras capture visual data and create detailed 3D maps of the lunar landscape, helping the rover identify obstacles, plan routes, and ensure safe traversal through challenging terrains.

Navigation Technology	Function and Application
Inertial Measurement Unit (IMU)	Detects the rover's acceleration and rotation, providing data for precise short-term navigation
Star Tracker	Locates the rover by comparing star positions with a known star map, enabling accurate absolute positioning
Terrain Mapping Cameras	Capture visual data and create detailed 3D maps of the lunar landscape, assisting in obstacle detection and navigation

Table 3: Navigation Technologies for Lunar Rover

To guarantee the mission's success, a combination of these navigational tools can be utilized. Each system has its own advantages and limitations. The star tracker is highly accurate for absolute positioning, but can face challenges in situations with limited visibility or when the stars are obstructed. The IMU can provide real-time data for short-term navigation, but it may drift over time, leading to positional errors. Visual odometry, based on the data from the terrain mapping cameras, allows the rover to estimate its motion by comparing consecutive images, but it can be affected by visual disturbances like lighting changes and surface texture variations. By integrating these different systems and employing sophisticated algorithms to fuse their data, the rover can overcome individual limitations and achieve robust and reliable navigation on the lunar surface. This comprehensive navigation system ensures that the rover efficiently accomplishes its scientific objectives and successfully explores the Moon.

Wheel Design

Since the Copernicus Crater is considered as the landing site, it is estimated that the lunar surface

is relatively not very inclined. Using this result, a wheel design that can climb to a maximum height of 0.2 m is considered. The rocker-bogie suspension was chosen because of its high efficiency for roving in various soil textures. It is known that the rocker-bogie suspension allows the rover to clear obstacles 1.5x the tire diameter [Efstratios et al., 2021]. According to the calculation, the minimum wheel diameter must be 0.133 m to work effectively.

The third parameter called drawbar pull is considered to compare the diameter and width. Drawbar pull is the force that is available to pull or push an extra weight until the maximum amount of traction is reached [Apostolopoulos, 2001]. It is the difference between traction and motion resistance. To calculate the drawbar pull, calculations from Equation 1 to Equation 6 are used with coefficients in the Moon condition. Equations in the article [Efstratios et al., 2021] were used as a reference for parameters that are not available in the literature.

For 0.14 m wheel diameter and 0.08 m wheel width; maximum sinkage calculated as 21.7 mm through;

$$z_{rw} = \frac{1}{N} \left(\frac{3W_W \cos \theta}{(3-n)(k_c + b_w k_\varphi) \sqrt{d_w}} \right)^{\frac{2}{(2n+1)}} \quad (1)$$

cohesive modulus	$k_c = 1400 \text{ N m}^{-2}$	frictional modulus	$k_\varphi = 8200 \text{ N m}^{-2}$
exponent of sinkage	$n = 0.8$	wheel width	$b_w = 0.08 \text{ m}$
wheel diameter	$n = 0.14 \text{ m}$	number of wheels	$N = 6$
cohesion	$c = 170 \text{ N m}^{-2}$	slope angle	$\theta = 30^\circ$
weight on moon	$W_W = 48.6 \text{ N}$		

$$H = c \left(\frac{\pi}{4} \right) \left(\sqrt{z_w d_w} \right) b_w + W_W \cos \theta \quad R_r = f_r W_W \quad (2)$$

$$R_c = \frac{1}{n+1} b_w \left(\frac{k_c}{b_w} + k_\varphi \right) z^{n+1} \quad R_g = \frac{W_w \sin \theta}{N} \quad (3)$$

$$DP = F - R_{all} = H - R_c - R_g - R_r \quad (4)$$

from these equations it is calculated that;

soil thrust	$H = 40.1 \text{ N}$	rolling resistance	$R_r = 2.43 \text{ N}$
soil compaction resistance	$R_c = 0.0993 \text{ N}$	gravitational resistance	$R_g = 4.65 \text{ N}$
drawbar pull(neglecting others)	$DP = 33.22 \text{ N}$		

where $z = 0.0055$ and friction coefficient $f_r = 0.05$

[Carrier et al., 1991; Efstratios et al., 2021]

According to the calculations, results are parametrized with 4 different diameters and 5 different width magnitudes, and Figure 12 is obtained. According to the figure, since there is no significant

difference in the drawbar pull compared to the width, a choice that will not increase the weight and volume and be durable should be made. Also, since the minimum wheel diameter is determined as 0.133m, it is necessary to evaluate the top two lines. When all the parameters are taken into account, the diameter range of 0.14-0.16m and the width range of 0.08-0.1m provides the necessary conditions. Therefore, 0.14m diameter and 0.08m with are chosen with the condition of making improvements in the next iterations.

Three different materials are considered for the wheel design: Aluminum, titanium, and reinforced carbon fiber rims with aluminum. According to the designed dimensions, aluminum wheels weight approximately 2.125 kg and titanium wheels weigh approximately 12 kg. Material selection and geometry can be optimized in further iterations.

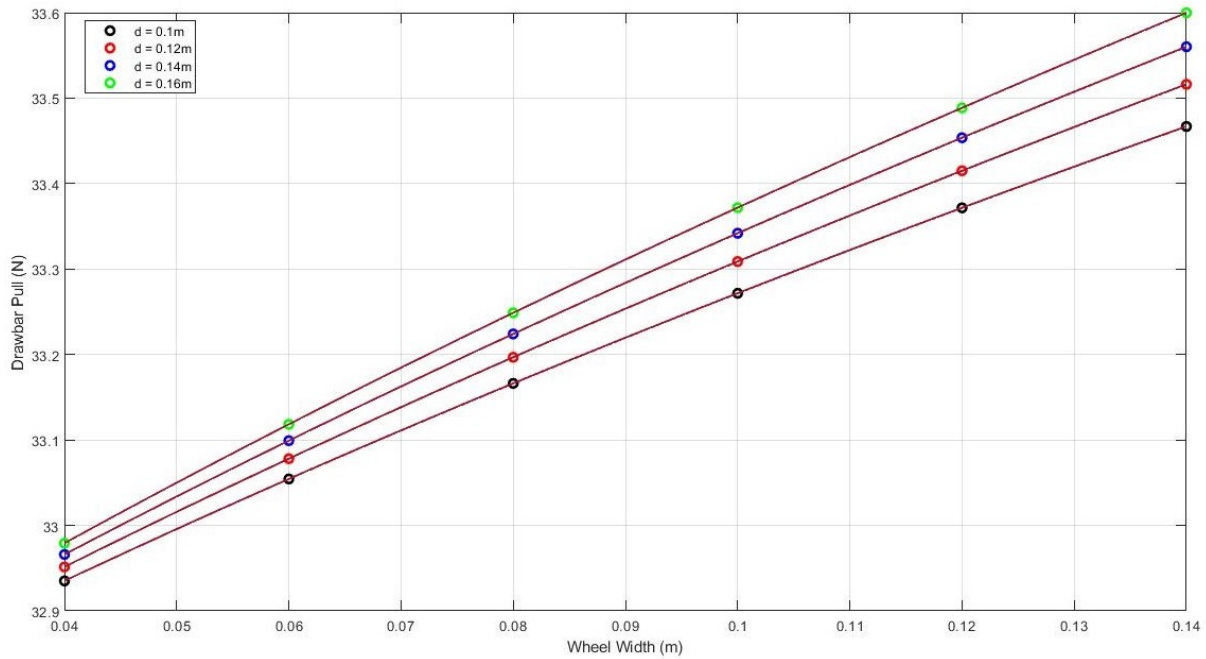


Figure 12: Drawbar Pull / Wheel Width for 4 different diameters

DATA HANDLING AND COMMUNICATION

Data Handling

The data handling system of the Turkish Lunar Rover plays a crucial role in collecting, processing, and storing the vast amounts of data gathered by its various scientific instruments. The data collected can encompass images, videos, measurements, and other essential scientific information.

Data Collection: To collect this diverse data, the rover is equipped with a range of specialized sensors and cameras. The sensors include a UV-VIS spectrophotometer and an X-ray fluorescence spectrometer, which analyze the composition of lunar materials and help identify different minerals present on the surface. Additionally, a variety of cameras, such as the panoramic camera, microscopic imager, and hazard detection cameras, provide valuable visual data from different angles and perspectives.

Data Processing and Storage: Once the data is collected, it undergoes necessary processing before being stored. The processing may involve converting raw data into a usable format and performing initial analysis. For example, image processing algorithms may be applied to enhance image quality

or to extract specific features for scientific study.

In terms of storage, the data handling system is equipped with high-capacity, reliable storage units capable of storing vast amounts of data gathered during the mission. Given the harsh lunar environment, data integrity, and security are paramount to ensure that valuable scientific data is not lost or corrupted.

Data Transmission Speeds: The data handling system is also responsible for data transmission back to the mission control on Earth. As the rover operates on the Moon, the communication link faces challenges, including the vast distance between Earth and the Moon. To overcome this, the rover's communication system is designed to have high data transmission speeds whenever a direct line of sight with Earth is established. Advanced communication protocols and high-gain antennas ensure efficient data transmission during these windows of opportunity. The speed of the communication may vary to the selections of electronics given in table 4. Furthermore, the rover may be programmed to prioritize and compress data, sending only critical scientific data first, followed by non-critical data during subsequent communication sessions.

Data Handling Component	Function and Description
Sensors and Cameras	Collection of various data, including temperature measurements, and visual imagery from different angles.
Data Processing Unit	Processing and conversion of raw data into usable formats, initial analysis, and feature extraction.
Data Transmission System	Facilitating high-speed data transmission to Earth during communication windows using advanced protocols and high-gain antennas.
Data Compression Algorithms	Compressing data to optimize transmission efficiency and prioritize critical scientific data during communication with Earth.

Table 4: Data Handling System Components

The data handling system of the Turkish Lunar Rover uses, comprising sensors, cameras, data processing units, and an advanced data transmission system, which are detailed in table 4. They are critical components that enable the successful execution of the rover's scientific mission on the Moon.

Communication

Communication systems are of paramount importance for transmitting data from the Turkish Lunar Rover to the ground station through the lander. The communication system can be divided into three crucial parts: the transceiver on the rover, the lander, and the ground station.

- **Rover:** The rover will be equipped with a transceiver and an antenna system designed for both data transmission and reception. It serves as the primary data collection platform on the lunar surface, acquiring scientific data, images, and telemetry. The rover's transceiver establishes communication links not only with the lander but also with Earth, ensuring that the data collected during lunar exploration is relayed back to mission control.
- **Lander:** The lander which carries the rover to the Moon's surface, serves as a vital intermediary in the communication chain. It will be equipped with its own transceiver and antenna, responsible for receiving data from the rover and transmitting it to the ground station on Earth. This relay function is essential due to the vast distance between the Moon and Earth, making direct communication between the rover and the ground station challenging.

- **Ground Station:** The ground station on Earth will consist of a high-gain antenna with transmitter and receiver systems. It will be able to send telecommand and receive the data transmitted by the lander, process it, and deliver it to the mission control center for analysis and decision-making. The ground station's capabilities are crucial for ensuring that the scientific data collected by the rover can be effectively utilized by the mission's scientific team.

Communication Method	Advantages	Disadvantages
Radio Waves	Reliability and ability to travel long distances	Lower data rates compared to other methods
Laser Communication	Higher data rates	Requires clear line of sight, and sensitive to atmospheric conditions
Satellite Communication	Continuous communication during periods with no clear line of sight between the lander and ground station	Slightly higher latency due to signal travel through satellites potential signal interference and complexity in integrating with satellite networks

Table 5: Communication Methods and Their Advantages/Disadvantages

By carefully considering the communication system options and implementing the appropriate methods, the Turkish Lunar Rover mission can ensure efficient and reliable data transmission, enabling valuable scientific exploration and successful execution of the mission objectives.

STRUCTURE AND MECHANISMS

In the Turkish Lunar Rover design, the size of the solar panels is planned as 45x65 cm and the wheels are 14 cm in diameter. Considering these parameters, the size of the rover has been calculated as 65x45x30 cm approximately and covered with solar panels on the top of the chassis.

Rocker bogie suspension is chosen to be implemented on the rover. The rocker-bogie suspension system ensures that all wheels of the rover are in contact with the ground, providing freedom of movement on rough terrain, and a better climbing performance. Although the trajectory of the rover is relatively flat, the Rocker bogie suspension is chosen for any unexpected terrain and environmental conditions. For the design, the suspension system is designed for a maximum of 20 cm obstacle clearance. DC brushless motor is used in each wheel so that each wheel can be steered independently of the other.

For the material selection, considering the radiation shielding character and weight, aluminum alloys are decided to use in many parts. In-wheel rims and at the points where heat transfer is not critical, it is planned to gain strength by reducing weight with carbon fiber alloys.

Considering all the parameters, it is predicted that the Turkish Lunar Rover will weigh approximately 30 kilograms when all the payloads and scientific experiment equipment are loaded. The mass is aimed to be limited to 30 kilograms by making the design upgrades. With more iterations in the future, it is aimed to reduce the weight and approach to the micro rover class.

POWER MANAGEMENT

The requirements of the power management of the rover mission are summarized as follows:

- Continuous power, must be provided over the full mission duration.
- Peak power needs of various scenarios must also be addressed.

Given the unique location of the Copernicus Crater and the duration of the mission, it is essential to have a reliable and efficient power source that can withstand the lunar environment. The power requirements for the rover have been estimated based on the hourly power usage of its various systems. To ensure a continuous power supply, a solar panel with high-efficiency photovoltaic cells will be used to harness solar energy from the lunar environment. Additionally, a non-rechargeable battery will be integrated into the power system, it will supply power as a backup to provide enough power to run the rover's systems and instruments for the entire mission duration in case of any issues with the solar panels, or when the generated power is not enough to run all systems. As an alternative to the charged batteries, Radioisotope Thermal Generators (RTGs) can be used to acquire needed power for the rover when the solar panels are defective. RTGs convert the thermal energy released by the radioactive material into electrical energy. Mostly plutonium-238 is used as fuel and in recent years, americium-241 which generates thermal power between 200-240 W is being developed by ESA as an alternative [Ambrosi et. al., 2019]. In space applications and in-situ research missions, RTG is mostly used for thermal management and sleep mode in the lunar night due to vast temperature changes. Developing a heat source consists of multiple layers to ensure safe launch of radioactive material. The system includes fuel, Pt-Rh cladding housing, and carbon composite aeroshell. Since the design and the application require detailed research and development process, it has been decided to use a battery design that has taken place in the literature instead of RTG.

Power Budget

To have a comprehensive understanding of the power requirements for different rover systems, the power consumption budget in full operation mode, scientific mode, and idle mode is provided in Table 6 where the hourly power consumption of each system is given in watts (W).

Subsystem	Full Operation Mode	Scientific Mode	Idle
Navigation and Mobility	12	0	0
Thermal System	2	2	2
Power Management	4	4	4
Communications	5	5	2
Processing	5	5	2
Imagery	9	9	4
Navigation System	8	0	0
Science Module	2	2	0
Total	47	27	14

Table 6: Power budget

Solar Irradiance and Power Generation

Orbital Positions		Total Solar Irradiance	
Earth	Moon	Near Side	Far Side
Perihelion	Perigee	1,401.65	1,415.54
Perihelion	Apogee	1,400.83	1,416.36
Aphelion	Perigee	1,311.18	1,323.74
Aphelion	Apogee	1,310.44	1,324.49

Table 7: Solar Irradiance due to the Revolution of Earth and Moon [Kaczmarzyk et al., 2018]

The solar irradiances presented in Table 7 indicate that the average solar irradiance on the near side of the Moon, where the Copernicus Crater is located, is approximately 1356 W/m^2 . Assuming a solar panel efficiency of 20% at the given temperatures and using solar panel sizes of $40\text{cm} \times 60\text{cm} = (0.24\text{m}^2)$, the estimated power output is calculated as follows:

$$\text{Power Output} = \text{Efficiency} \times \text{Solar Irradiance} \times \text{Solar Panel Area}$$

$$\text{Power Output} = 0.2 \times 1356 \text{ W/m}^2 \times 0.24 \text{ m}^2 = 65 \text{ W}$$

It's important to consider that this solar irradiance value was average, and therefore the actual power produced could vary within a certain range. The final power output is estimated to be $65 \pm 4\% \text{ W}$, which is greater than our estimated required power.

The systems will use approximately 40W on average. In the worst case scenario, for the rover to function for 12 days as backup, the rover will need:

$$\text{Total Power Needed} = 40 \text{ W/h} \times 24 \text{ h/d} \times 12 \text{ d} = 12.520 \text{ W}$$

Which can be provided by charged batteries.

Power Management and Efficiency Strategies

To ensure optimal power management during the mission, the rover's control software will be designed to minimize power consumption whenever possible. The power management system will continuously monitor the energy levels, system requirements, and operational modes to balance power usage effectively. Some power-saving strategies that will be implemented include:

- **Sleep Mode:** The rover will enter a low-power sleep mode during idle periods, reducing power consumption to a minimum.
- **Dynamic Power Allocation:** Power distribution among different systems will be dynamically adjusted based on their real-time demands.
- **Sensor Control:** Sensors will be activated only when necessary, conserving power and extending their operational life.
- **Communication Protocol:** Optimized communication protocols will be used to minimize data transmission power.

By implementing these power management and efficiency strategies, the mission's power requirements can be met while ensuring the rover's continuous and reliable operation throughout its lunar exploration mission at the Copernicus Crater.

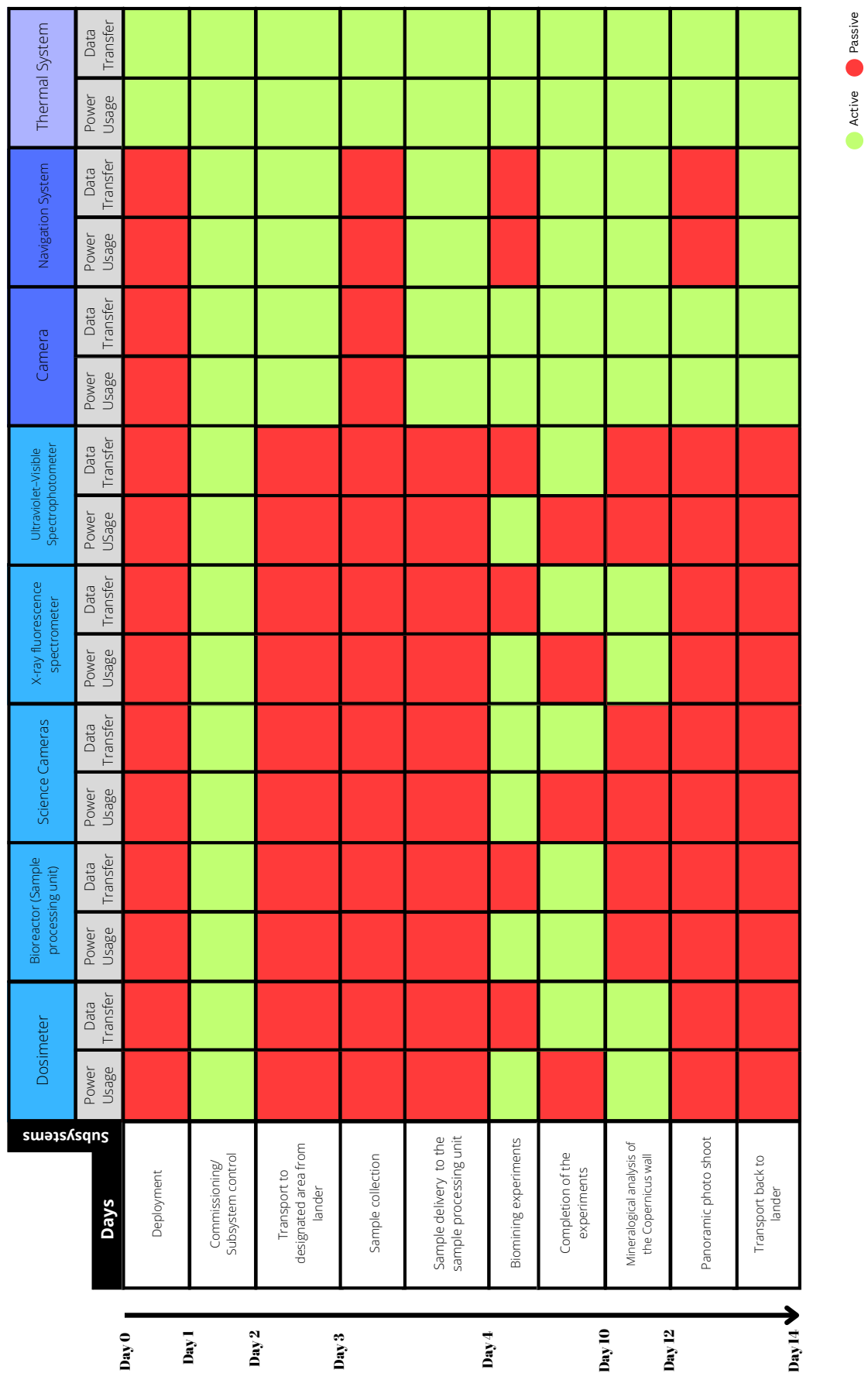


Figure 13: Mission Timeline

THERMAL MANAGEMENT

Since there is no atmosphere around the Moon, drastic temperature changes are observed on the lunar surface. The temperature difference between day and night can reach almost 250°C. That is why the importance of the thermal design is significant for the design. In the Turkish lunar mission rover case, a mission plan is designed for around half a Lunar day (10-12 Earth days) time. For this reason, the thermal design is focused on adapting to the warm environment rather than the cold environment and providing an internal temperature balance for electronic device operations. As indicated by the article [Oikawa et al., 2016], at 45° latitude on the moon, the temperature rises from -87.9°C at 0.5 days after sunrise, up to 81.5°C during midday. In the Turkish lunar mission rover case, the center of the Copernicus Crater is located 8.314°N. Due to the low axial tilt of the moon and the Copernicus Crater's location, which is closer to the equator, it is predicted that larger temperatures will be encountered compared to latitudes around 45°. Therefore, in further studies, these conditions should be considered while performing thermal modeling. Certain mission scenarios shall be studied while considering solar radiation, surface albedo, infrared radiation and radiation to outer space for interactions with external environment as well as thermal conduction and internal heat dissipation for internal interactions.

To perform with maximum efficiency, each electronic device operates in a temperature range. Chassis design in accordance with the temperature range is crucial for the mission operation. Therefore, to investigate the temperature range, a sample of electronic equipment is considered from CubeSat hardware and tabulated. According to Table 8, it is important to make a thermal design of the chassis in the range of (-5, +45)°C which is the range of the battery in charge.

Equipment	Min	Max
Motor	-40	100
Motor Controller	-20	70
Cameras	-40	85
CPU	-20	125
IMU	-40	100
Battery in Discharge	-20	60
Radio	-40	85
Power Distribution Unit	-40	85
Ethernet Switch	-40	60

Table 8: CubeSat Equipment Temperatures

Thermal management has an important role in the efficiency and lifetime of the components. That is why, several methods to manage thermal distribution are suggested. One of the suggestions for thermal management is composite material integration. For example, carbon fiber provides a great reduction in weight compared to aluminum, while conducting less heat. This limits the use of carbon fiber in areas where heat dissipation is required. Another option for the management is using heat dissipators such as heat sinks and radiators. By constructing fluid loops, excess heat in electronics or moving parts can be dissipated with a fluid medium. Moreover, double-acting Peltier elements that work both for heating and cooling can be another option for thermoelectric heat management. In case of the need for a heating system, radioisotope heater units (RHU) can be used due to their high efficiency in their lifetime by using radioactive materials. Finally, insulation materials are planned to be used for the major part of thermal management. Multi-layer blankets are considered to protect the rover from the Sun's radiation and composite insulators are considered to prevent conductive heat transfer by separating the parts that must be at different temperatures such as solar panels and chassis.

GENERAL ROVER SYSTEM DESCRIPTION

The product tree of the conceptual rover design is provided in Figure 13.

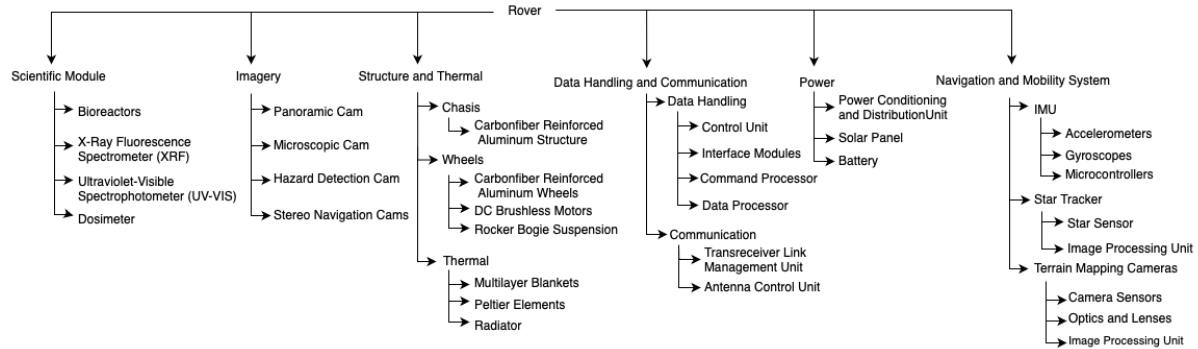


Figure 13: Product Tree

As it can be seen from the product tree, there are six subsystems to realize the designed rover mission.

CONCLUSION

In this paper, a conceptual rover design study was presented for a future Lunar surface exploration mission. The rover mission and scientific experiments together with the associated landing site and trajectory design were proposed. The subsystems necessary to support such mission were also analyzed. It is intended that the mission design and technical discussions presented in this paper will assist the upcoming Turkish Lunar soft landing mission.

ACKNOWLEDGMENTS

We would like to thank Adj. Prof. Luis Zea for his comments and inputs on the scientific aspects of the rover mission.

Also special thanks to Zeynep Altay and Sena Nur Çalışkan for their inputs in terms of the space environment and utilization of resources in potential landing sites.

References

- Abhilash, Mehta, K. D., Kumar, V., Pandey, B. D., & Tamrakar, P. K. (2010). *Column bioleaching of a low-grade silicate ore of uranium*. Mineral Processing and Extractive Metallurgy Review, 31(4), 224–235. <https://doi.org/10.1080/08827508.2010.483380>
- Ambrosi, R.M., Williams, H., Watkinson, E.J. et al. (2019). *European Radioisotope Thermoelectric Generators (RTGs) and Radioisotope Heater Units (RHUs) for Space Science and Exploration*. Space Sci Rev 215, 55. <https://doi.org/10.1007/s11214-019-0623-9>
- Apostolopoulos, D. (2001). *Analytical Configuration of Wheeled Robotic Locomotion*. The Robotics Institute Carnegie Mellon University
- Arai, T., Ohtake, M., Yamamoto, A., Sugihara, T., Hiroi, T., Nakamura, R., Namiki, N., Wada, K., Yamamoto, S., Matsunaga, T., & Haruyama, J. (2011). *Possible Crustal Boundary Exposed at Lunar Copernicus Crater*.
- Averesch, N. J., Berliner, A. J., Nangle, S. N., Zezulka, S., Vengerova, G. L., Ho, D., Casale, C. A., Lehner, B. A., Snyder, J. E., Clark, K. B., Dartnell, L. R., Criddle, C. S., & Arkin, A. P. (2023). *Microbial biomanufacturing for space-exploration—what to take and when to make*. Nature Communications, 14(1). <https://doi.org/10.1038/s41467-023-37910-1>
- Baygeldi, A., Bayram, İ., Pehlivan, B., Ayvaci, B., "Moon Research Program of Türkiye". IEEE Xplore/RAST 2023, Istanbul, Türkiye, 7-9 June 2023.
- Biorock - Extraterrestrial Geomicrobiological Package Test Bed for the ISS. EEA - European Space Agency. (n.d.). <https://eea.spaceflight.esa.int/portal/exp/?id=9693>
- Bogue, R.(2012). *Robots for space exploration*. Industrial Robot: An International Journal. 39. <https://doi.org/10.1108/01439911211227872>.
- Carrier, D., Olhoeft, G., Mendell, W. (1991). *Lunar Source Book: A User's Guide to the Moon*. Cambridge: Cambridge Univ. Press, 475
- Chaudhury, G. R. (2010). *Bioleaching of low-grade uranium ore using Acidithiobacillus ferrooxidans*. Indian Journal of Microbiology, 50(1), 70–75. <https://doi.org/10.1007/s12088-010-0015-z>
- Cockell, C. S., Santomartino, R., Finster, K., Waajen, A. C., Eades, L. J., Moeller, R., Rettberg, P., Fuchs, F. M., Van Houdt, R., Leys, N., Coninx, I., Hatton, J., Parmitano, L., Krause, J., Koehler, A., Caplin, N., Zuijderduijn, L., Mariani, A., Pellari, S. S., ... Demets, R. (2020). *Space station biomineralization experiment demonstrates rare earth element extraction in microgravity and Mars Gravity*. Nature Communications, 11(1). <https://doi.org/10.1038/s41467-020-19276-w>
- de Vera, J.-P., Boettger, U., Noetzel, R. de, Sánchez, F. J., Grunow, D., Schmitz, N., Lange, C., Hübers, H.-W., Billi, D., Baqué, M., Rettberg, P., Rabbow, E., Reitz, G., Berger, T., Möller, R., Bohmeier, M., Horneck, G., Westall, F., Jänicke, J., ... Spohn, T. (2012). *Supporting mars exploration: BIOMEX in low Earth orbit and further astrobiological studies on the Moon using Raman and PanCam technology*. Planetary and Space Science, 74(1), 103–110. <https://doi.org/10.1016/j.pss.2012.06.010>
- Desouky, O. A., El-Mougith, A. A., Hassanien, W. A., Awadalla, G. S., & Hussien, S. S. (2016). *Extraction of some strategic elements from thorium–uranium concentrate using bio-products of aspergillus ficuum and pseudomonas aeruginosa*. Arabian Journal of Chemistry, 9. <https://doi.org/10.1016/j.arabjc.2011.08.010>

- Domańska, M., Hamal, K., Jasionowski, B., & Łomotowski, J. (2019). *Bacteriological contamination detection in water and wastewater samples using OD600*. Polish Journal of Environmental Studies, 28(6), 4503–4509. <https://doi.org/10.15244/pjoes/94838>
- Dhara, S., & Misra, N. L. (2019). *Elemental characterization of nuclear materials using total reflection X-ray fluorescence spectrometry*. TrAC Trends in Analytical Chemistry, 116, 31–43. <https://doi.org/10.1016/j.trac.2019.04.017>
- Fogtman, A., Baatout, S., Baselet, B., Berger, T., Hellweg, C. E., Jiggins, P., La Tessa, C., Narici, L., Nieminen, P., Sabatier, L., Santin, G., Schneider, U., Straube, U., Tabury, K., Tinganelli, W., Walsh, L., & Durante, M. (2023). *Towards sustainable human space exploration—priorities for radiation research to quantify and mitigate radiation risks*. Npj Microgravity, 9(1). <https://doi.org/10.1038/s41526-023-00262-7>
- Gumulya, Y., Zea, L. and Kaksonen, A.H. (2022). *In situ resource utilisation: The potential for space biomining*. Minerals Engineering, 176, p. 107288. doi:10.1016/j.mineng.2021.107288.
- Horneck, G. (2011) *Lithopanspermia*, Encyclopedia of Astrobiology, 931–931. https://doi.org/10.1007/978-3-642-11274-4_899
- Jerez, C. A. (2017). *Metal extraction and biomining. Reference Module in Life Sciences*. <https://doi.org/10.1016/b978-0-12-809633-8.13077-8>
- Jolliff, B. L., Gillis, J. J., Haskin, L. A., Korotev, R. L., & Wieczorek, M. A. (2000). *Major lunar crustal terranes: Surface expressions and crust-mantle origins*. Journal of Geophysical Research: Planets, 105(E2), 4197–4216. <https://doi.org/10.1029/1999je001103>
- Jones, A. (2019). *China grew two leaves on the Moon: The Chang'E-4 spacecraft also carried potato seeds and fruit-fly eggs to the lunar far side*. IEEE Spectrum, 56(11), 9–10. <https://doi.org/10.1109/mspec.2019.8889900>
- HASSANIEN, W. A. G., DESOUKY, O. A. N., & HUSSIEN, S. S. E. (2013). *Bioleaching of some Rare Earth Elements from Egyptian Monazite using Aspergillus ficuum and Pseudomonas aeruginosa*. Walailak Journal of Science and Technology (WJST), 11(9), 809–823. Retrieved from <https://wjst.wu.ac.th/index.php/wjst/article/view/481>
- Kaczmarzyk, Marcin, Gawronski, Marcin, & Piatkowski, Grzegorz. (2018). *Global database of direct solar radiation at the Moon's surface for lunar engineering purposes*. E3S Web Conf. 49 00053 <https://doi.org/10.1051/e3sconf/20184900053>
- Korotev, R. L., Jolliff, B. L., Zeigler, R. A., Seddio, S. M., & Haskin, L. A. (2011). *Apollo 12 revisited*. Geochimica et Cosmochimica Acta, 75(6), 1540–1573. <https://doi.org/10.1016/j.gca.2010.12.018>
- Lefimil, C., Osorio, H., Quatrini, R., Holmes, D. S., & Jedlicki, E. (2009). *Regulation of expression of the PetI operon involved in iron oxidation in the biomining bacterium Acidithiobacillus ferrooxidans*. Advanced Materials Research, 71–73, 199–202. <https://doi.org/10.4028/www.scientific.net/amr.71-73.199>
- Li, R., Guo, X., Xiong, K., Xing, X., & Zhang, C. (2023). *Microbial micro-tube culture system: A miniature bioreactor for controllable bubble-free oxygen supply based on high gas-permeability teflon tube*. Biochemical Engineering Journal, 191, 108789. <https://doi.org/10.1016/j.bej.2022.108789>
- Loudon, C.-M., Nicholson, N., Finster, K., Leys, N., Byloos, B., Van Houdt, R., Retberg, P., Moeller, R., Fuchs, F. M., Demets, R., Krause, J., Vukich, M., Mariani,

- A., & Cockell, C. (2017). *Biorock: New experiments and hardware to investigate microbe-mineral interactions in space*. *International Journal of Astrobiology*, 17(4), 303–313. <https://doi.org/10.1017/s1473550417000234>
- Lucey, P. G., Hawke, B. R., & Horton, K. (1991). *The distribution of olivine in the crater Copernicus.*, *Geophysical Research Letters*, 18(11), 2133–2136. <https://doi.org/10.1029/91gl02538>
- Mathiyazhagan, N. (2011). *Bioremediation on effluents from magnesite and bauxite mines using thiobacillus SPP and pseudomonas spp.* *Journal of Bioremediation & Biodegradation*, 02(01). <https://doi.org/10.4172/2155-6199.1000115>
- Oberbeck, V. R. (1975). *The role of ballistic erosion and sedimentation in lunar stratigraphy*. *Reviews of Geophysics*, 13(2), 337. <https://doi.org/10.1029/rg013i002p00337>
- Oikawa, T., Tanaka, T., Walker1, J., Uno, K., Costa, P., Britton, N., Yoshida, K. (2016). *Thermal Design and Analysis of Conceptual Flight Model for a Lunar Exploration Rover*.
- Pal, S., Pradhan, D., Das, T., Sukla, L. B., & Chaudhury, G. R. (2010). *Bioleaching of low-grade uranium ore using Acidithiobacillus ferrooxidans*. *Indian Journal of Microbiology*, 50(1), 70–75. <https://doi.org/10.1007/s12088-010-0015-z>
- Qiong W., Jizhong L. (2016). *TA Chang'e-4 mission concept and vision of future Chinese lunar exploration activities*. *Acta Astronautica*, 127, 678-683,0094-5765. <https://doi.org/10.1016/j.actaastro.2016.06.024>.
- Quatrini, R., Jedlicki, E., & Holmes, D. S. (2005). *Genomic insights into the iron uptake mechanisms of the biomining microorganism acidithiobacillus ferrooxidans*. *Journal of Industrial Microbiology. Biotechnology*, 32(11–12), 606–614. <https://doi.org/10.1007/s10295-005-0233-2>
- Reisz, J. A., Bansal, N., Qian, J., Zhao, W., & Furdui, C. M. (2014). *Effects of ionizing radiation on biological molecules—mechanisms of damage and emerging methods of detection*. *Antioxidants & Redox Signaling*, 21(2), 260–292. <https://doi.org/10.1089/ars.2013.5489>
- Reitz, G. (2008). *Characteristic of the radiation field in low Earth orbit and in Deep Space*. *Zeitschrift Für Medizinische Physik*, 18(4), 233–243. <https://doi.org/10.1016/j.zemedi.2008.06.015>
- Rigas, E., Kostopoulos, V. (2021). *Design of a Mars exploration microrover concept for future missions with modularity and reconfigurability characteristics*. 72nd International Astronautical Congress (IAC)
- Rocha, F. S., Gomes, A. J., Lunardi, C. N., Kaliaguine, S., & Patience, G. S. (2018). *Experimental methods in chemical engineering: Ultraviolet visible spectroscopy-UV-vis*. *The Canadian Journal of Chemical Engineering*, 96(12), 2512–2517. <https://doi.org/10.1002/cjce.23344>
- Schramm, R. (2016). *Use of X-ray fluorescence analysis for the determination of rare earth elements*. *Physical Sciences Reviews*, 1(9). <https://doi.org/10.1515/psr-2016-0061>
- Scott, D., Andrews, J., & Lozano, P. (2021). *In Lunar Science: Advances and Opportunities (pp. 109-131)*. *Lunar Rover Camera Systems*. Springer. https://doi.org/10.1007/978-3-030-71798-9_5
- Sezonov, G., Joseleau-Petit, D., & D'Ari, R. (2007). *Escherichia coli Physiology in Luria-Bertani Broth*. *Journal of Bacteriology*, 189(23), 8746–8749. <https://doi.org/10.1128/jb.01368-07>
- Shahrubudin, N., Lee, T. C., & Ramlan, R. (2019). *An overview on 3D printing technology: Technological, materials, and applications*. *Procedia Manufacturing*, 35, 1286–1296. <https://doi.org/10.1016/j.promfg.2019.06.089>

- Siddiqi, A. A. (2018) *Beyond Earth A chronicle of deep space exploration, 1958–2016*, <https://www.nasa.gov/sites/default/files/atoms/files/beyond-earth-tagged.pdf>
- Sreenivasulu, N., & Fernie, A. R. (2022) *Diversity: Current and prospective secondary metabolites for nutrition and medicine*, *Current Opinion in Biotechnology*, 74, 164–170. <https://doi.org/10.1016/j.copbio.2021.11.010>
- Sun, Z., Jia, Y & Zhang, H. (2013). *Technological advancements and promotion roles of Chang'e-3 lunar probe mission*, *Science China Technological Sciences*. 56. 2702-2708. <https://doi.org/10.1007/s11431-013-5377-0>.
- Tang, J., Jiang S., Chen, C., Quan, Q., Yuan, F. & Deng, Z. (2016). *Drilling States Monitoring for a Planetary Drilling & Coring Testbed (PDCT): Method and Design*, *IEEE International Conference on Robotics and Biomimetics (ROBIO)*, Qingdao, China, 179-184. <https://doi.org/10.1007/s11431-013-5377-0>.
- Tang, J., Yang, T., Chen X., Zhang, Z., Tian, Y., Zhang, W. & Jiang, S. (2022). *Mechanical Characteristics of Lunar Regolith Drilling and Coring and Its Crawling Phenomenon: Analysis and Validation*, *Aerospace* 2022, 9, 709. <https://doi.org/10.3390/aerospace9110709>
- Yağlıoğlu, B., İnaltekin, B. B., Gökten, M., "The First Turkish Lunar Mission Part 1: Programmatic, Mission and System Aspects", *IEEE Xplore/RAST 2023*, Istanbul, Türkiye, 7-9 June 2023.
- Yamashita, N., Hasebe, N., Reedy, R. C., Kobayashi, S., Karouji, Y., Hareyama, M., Shibamura, E., Kobayashi, M.-N., Okudaira, O., d'Uston, C., Gasnault, O., Forni, O., & Kim, K. J. (2010). *Uranium on the Moon: Global Distribution and u/th ratio* *Geophysical Research Letters*, 37(10). <https://doi.org/10.1029/2010gl043061>
- Young, A. (2017). *Apollo Lunar Landing Missions 11, 12, and 14. The Apollo Lunar Samples*, 45–66. https://doi.org/10.1007/978-1-4614-6185-2_4
- Walker, J., Burtz, L., Dubois, F., Calzada-Diaz, A., Tanaka, T. (2019). *ispace Modular Rover Design* & *Biotechnology*, 32(11–12), 606–614. 70th International Astronautical Congress (IAC), 21-25.
- Walther, I. (2002). *Space bioreactors and their applications. Cell Biology and Biotechnology in Space*, 197–213. [https://doi.org/10.1016/s1569-2574\(02\)08020-6](https://doi.org/10.1016/s1569-2574(02)08020-6)